

CO4 - AT1 - LEVEL 2 Exam

Level Exam C 02 Sep 2026, 10:00 AM

Back

Result Available
Your score: 65 / 100

QUESTIONS

LEVEL 2 — MEDIUM

Q1

1DArray

Q2

structure

Q3

string

Q4

2DArray

2 / 4 saved

Level 2 — Medium union of two arrays (M) — 1DArray

Write a program to find the union of two arrays.

Sample Input:

Enter number of elements in first array: 4

Enter elements: 1 3 4 5

Enter number of elements in second array: 5

Enter elements: 2 3 5 7 8

Sample Output:

Union of arrays: 1 2 3 4 5 7 8

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {1, 3, 4, 5};
5     int b[] = {2, 3, 5, 7, 8};
6     int n = sizeof(a) / sizeof(a[0]);
7     int m = sizeof(b) / sizeof(b[0]);
8     printf("Enter number of elements in the first array");
9     printf("\nEnter elements:", a);
10    printf("\nEnter numbers of elements in the second array");
11    printf("\nEnter elements:", b);
```

Program Input (stdin)

Enter input if required...

Output

```
Enter number of elements in the first
array
Enter elements:
Enter numbers of elements in the second
array
```

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2 / 4 saved

```

1 #include <stdio.h>
2
3 int main() {
4     int a[] = {1, 3, 4, 5};
5     int b[] = {2, 3, 5, 7, 8};
6     int n = sizeof(a) / sizeof(a[0]);
7     int m = sizeof(b) / sizeof(b[0]);
8     printf("Enter number of elements in the first Array");
9     printf("\nEnter elements: ", a);
10    printf("\nEnter numbers of elements in the second Array");
11    printf("\nEnter elements: ", b);
12    printf("\nThe union of Arrays: ");
13
14    int i = 0, j = 0, k = 0;
15    int c[n + m];
16
17    while (i < n && j < m) {
18        int w;
19        if (a[i] < b[j]) {
20            w = a[i++];
21        } else if (b[j] < a[i]) {
22            w = b[j++];
23        } else {
24            w = a[i];
25            i++;
26            j++;
27        }
28        if (k == 0 || c[k - 1] != w) {
29            c[k++] = w;
30        }
31    }
32    while (i < n) {
33        int w = a[i++];
34        if (k == 0 || c[k - 1] != w) {
35            c[k++] = w;
36        }
37    }
38    while (j < m) {
39        int w = b[j++];
40        if (k == 0 || c[k - 1] != w) {
41            c[k++] = w;
42        }
43    }
44    for (int t = 0; t < k; t++) {
45        printf("%d ", c[t]);
46    }
47
48    return 0;
49 }

```

Enter input if required...

Output

```

Enter number of elements in the first
Array
Enter elements:
Enter numbers of elements in the second
Array

```

Run Save

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C PROGRAMMING

2 / 4 saved

```

1 #include <stdlib.h>
2
3 int main() {
4     int n, i, j;
5     printf("Enter number of elements: ");
6     if (scanf("%d", &n) != 1) return 0;
7     int *arr = (int *)malloc(n * sizeof(int));
8     printf("Enter elements: ");
9     for (i = 0; i < n; i++) {
10        scanf("%d", &arr[i]);
11    }
12    int smallest_missing = 1;
13    while (1) {
14        int found = 0;
15        for (i = 0; i < n; i++) {
16            if (arr[i] == smallest_missing) {
17                found = 1;
18                break;
19            }
20        }
21        if (!found) {
22            break;
23        }
24        smallest_missing++;
25    }
26    printf("First missing positive integer: %d\n", smallest_missing);
27    free(arr);
28    return 0;
29 }

```

Enter input if required...

Output

```

Enter number of elements: Enter elements:
First missing positive integer: 2

```

Run Save